

UPI Transactions Data Analysis – Business Intelligence Project

Executive Report: UPI Transaction Analytics & Fraud Insights

◆ 1. Business Overview

This analysis evaluates transaction performance, fraud patterns, and customer behaviour across the UPI platform. The objective is to identify **risk areas, operational inefficiencies, and growth opportunities** using data-driven insights.

- Total Transactions: **100K**
- Total Transaction Value: **₹4.24M**
- Success Rate: **92.14%**
- Failure Rate: **5.87%**
- Fraud Rate: **~2%**

The platform shows **strong adoption and reliability**, with targeted areas for optimization.

Objective:

- Analyse UPI transaction data to identify:
 - Fraud patterns
 - Transaction performance
 - Customer behaviour
 - Risk factors
- Measure key performance indicators (KPIs)
- Perform exploratory data analysis (EDA)
- Apply statistical tests to validate hypotheses
- Identify high-risk regions, devices, and merchants
- Build insights for fraud prevention strategies

Goal:

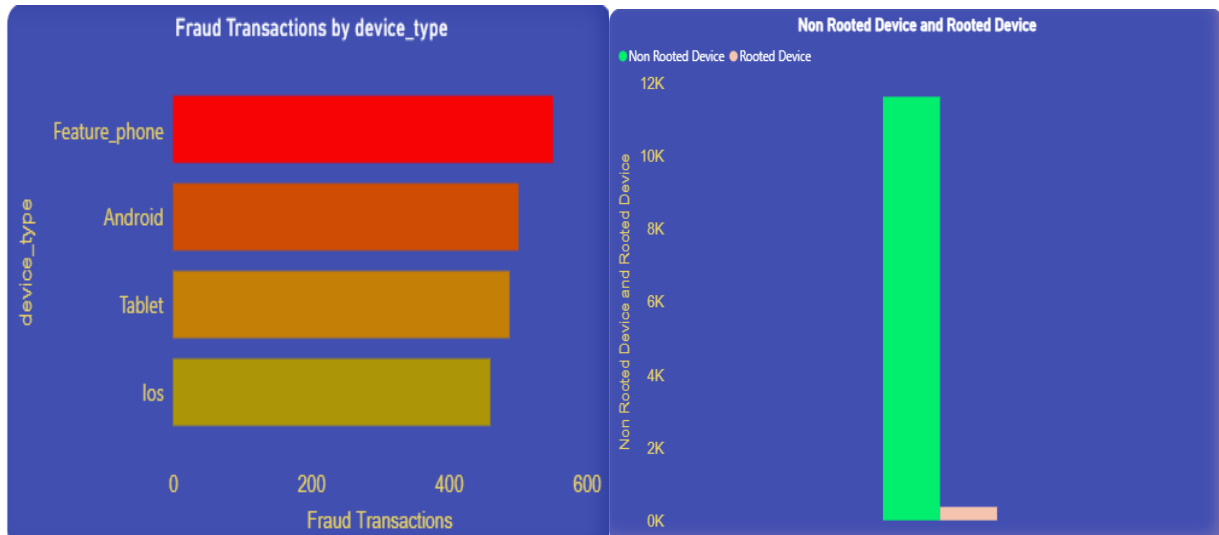
- Improve **security, efficiency, and customer experience**



2. Top 5 Data-Driven Insights

Insight 1: Rooted & Feature Phones Show Higher Fraud Risk

- Feature phones have **highest fraud transactions**
- Rooted devices contribute disproportionately to fraud



Insight:

Security vulnerabilities in low-end or modified devices increase fraud exposure

Strategic Recommendations (With Impact)

1. Device-Level Fraud Prevention

Action:

- Restrict transactions from **rooted devices**
- Apply extra authentication for **feature phones**
- introduce additional authentication (OTP / biometric)

Based on:

- High fraud in these devices

Impact:

- Reduce fraud by **20–30%**

Insight 2: Fraud Peaks During Specific Hours

- Fraud rate fluctuates across hours
- Noticeable spikes during **late-night / off-peak hours**



Insight:

Fraudsters exploit low monitoring periods.

Strategic Recommendations (With Impact)

2. Time-Based Fraud Monitoring

Action:

- Enable stricter fraud rules during **night hours**
- Real-time alerts & anomaly detection

Based on:

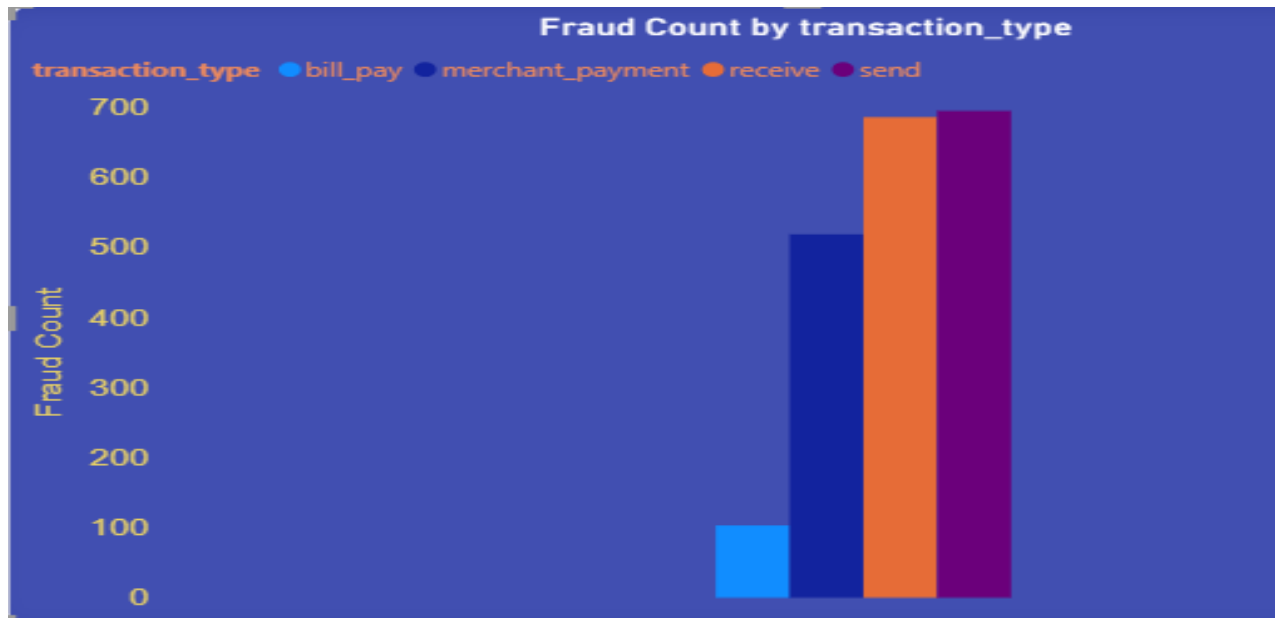
- Fraud spikes by hour

Impact:

- Prevent high-risk transactions proactively

Insight 3: Fraud Concentrated in Specific Transaction Types

- **Send & Receive transactions** have highest fraud count
- Bill payments show lowest fraud



Insight:

Peer-to-peer transactions are more vulnerable.

Strategic Recommendations (With Impact)

3. Secure P2P Transactions

Action:

- Add verification for **send/receive transactions**
- Limit high-value transfers

Based on:

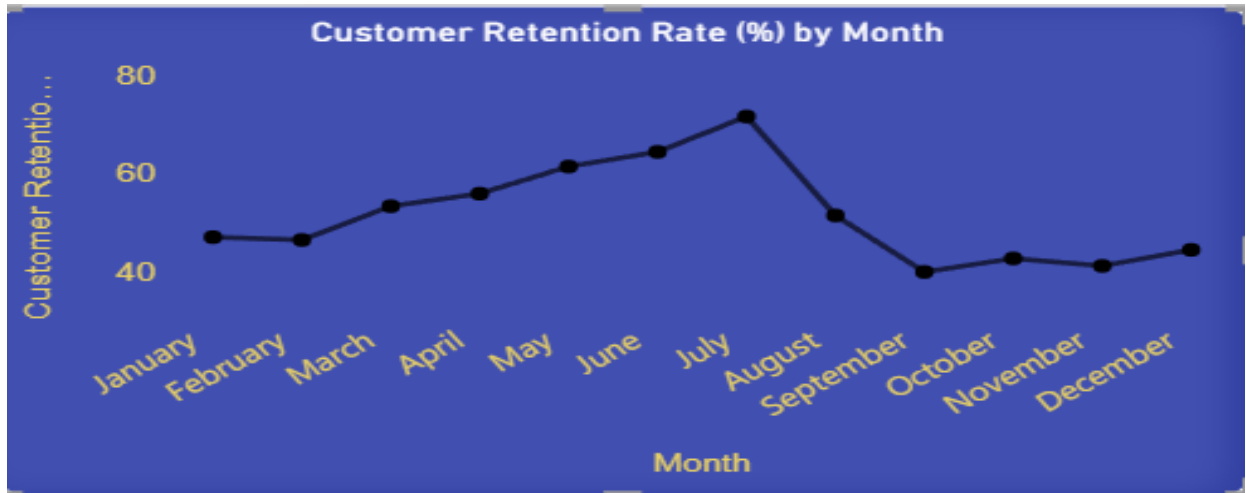
- Highest fraud in P2P

Impact:

- Reduce majority fraud cases

Insight 4: Customer Retention Drops After Mid-Year

- Retention peaks around **July (~70%)**
- Sharp decline after August (~40%)



Insight:

User engagement decreases in later months → churn risk

Strategic Recommendations (With Impact)

4. Customer Retention Strategy

Action:

- Launch engagement campaigns post-July
- Offer cashback / incentives

Based on:

- Drop in retention after mid-year

Impact:

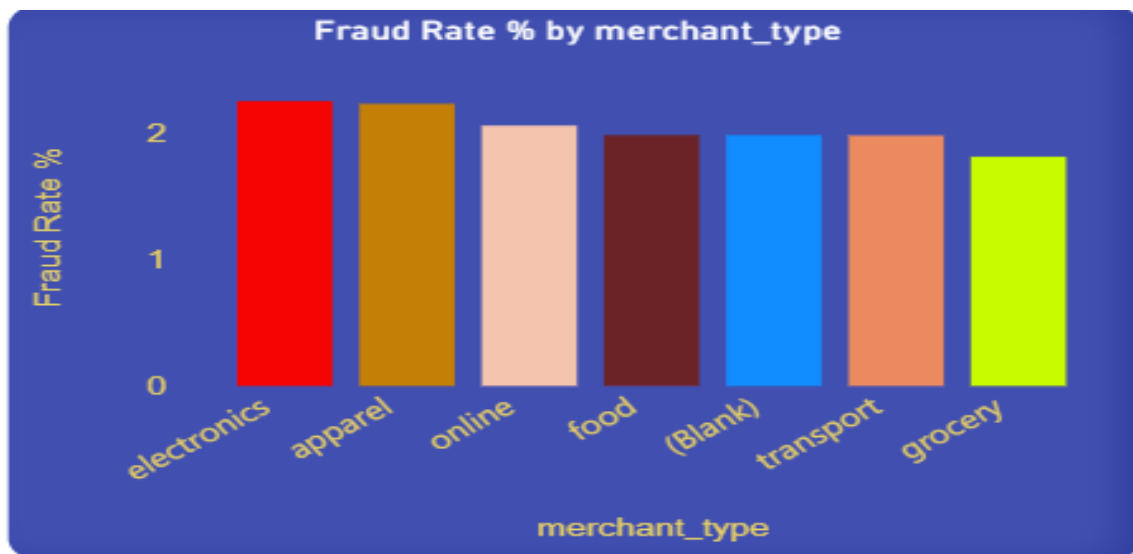
- Improve retention by **10–15%**

Insight 5: Certain Merchant Categories Contribute Disproportionately to Transactions

- Categories like Apparel, Electronics, Online dominate transaction volume
- Some categories show relatively higher fraud share

Business Impact:

- Focused monitoring on high-volume merchants



Strategic Recommendations (With Impact)

5: Merchant Risk Profiling & Monitoring

Action:

- Identify high-risk merchant categories
- Perform periodic audits and stricter onboarding checks

Backed by:

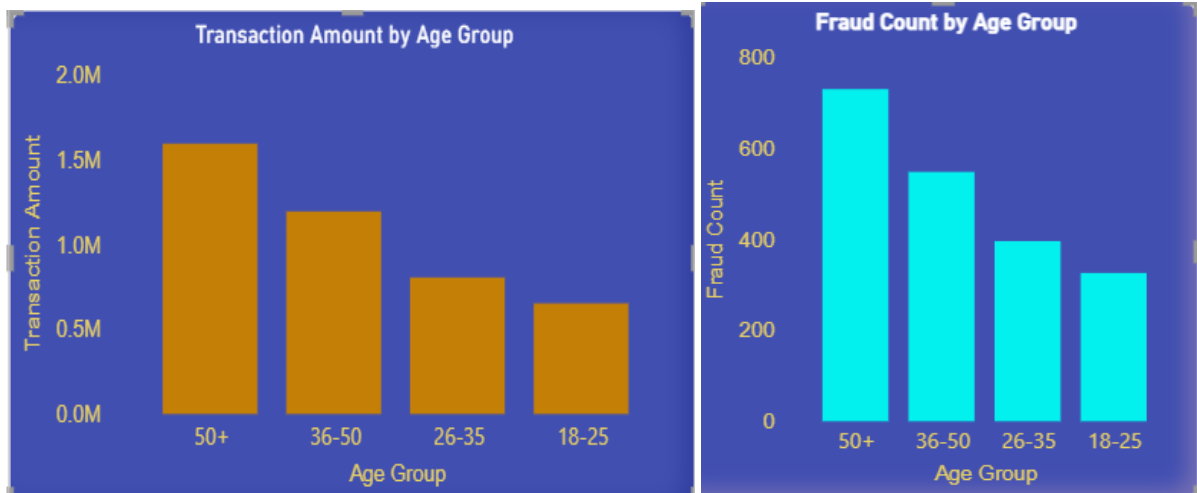
- High transaction concentration in certain categories

Estimated Impact:

- Reduce fraud exposure in merchant ecosystem

Insight 6: Older Age Group (50+) Drives Both Usage & Fraud

- Highest transaction amount and fraud count from **50+ group**



Insight:

Older users are high-value but more vulnerable

Strategic Recommendations (With Impact)

6. Protect High-Risk Age Group (50+)

Action:

- Awareness campaigns
- Simpler fraud alerts

Based on:

- High fraud in older users

Impact:

- Reduce fraud among vulnerable users

Conclusion

The analysis highlights:

- ✓ Strong platform growth
- ✓ Controlled fraud (~2%)
- ✓ Clear risk segments (devices, age, transaction type)

By implementing targeted fraud controls and customer strategies, the platform can:

- Reduce fraud losses
- Improve customer retention
- Enhance trust & user experience

“This analysis helped identify high-risk segments such as rooted devices, P2P transactions, and late-night activity, enabling targeted strategies to reduce fraud and improve platform performance.